



United International University

School of Science and Engineering

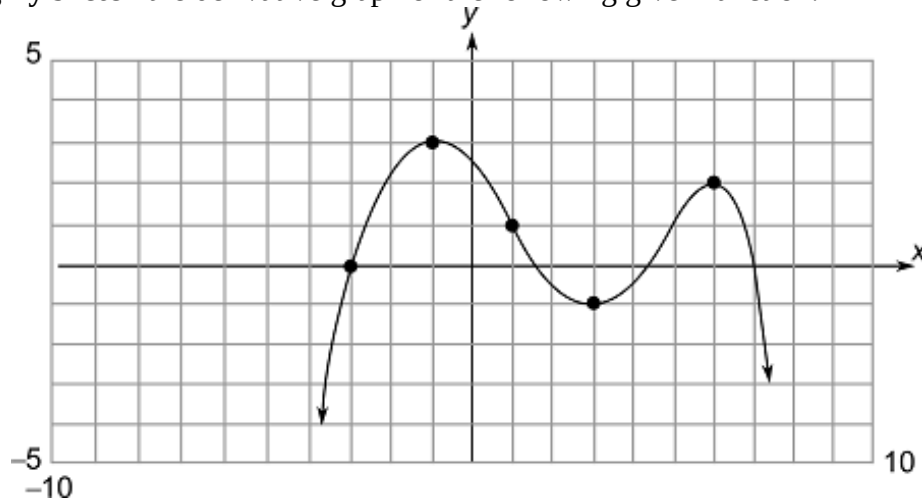
Final Examination Trimester: Fall-2023

Course Title: Fundamental Calculus

Course Code: Math 1151 Marks: 40 Time: 2 Hours

Answer all the questions. Answer all parts of a question together.

1. Consider the function $f(x) = x^2 - 2x$; $x_0 = -1$ and $x_1 = 2$.
- (a) Find the instantaneous rate of change of $f(x)$ with respect to x at an arbitrary value of x_0 . [2]
 - (b) Use part (a) to find the slope of the tangent lines for the value of x_0 . [1]
 - (c) Find the average rate of change of function in the interval $[x_0, x_1]$. [1]
 - (d) Find the equation of the tangent line to the function $f(x)$ at x_0 . [2]
 - (e) Find the equation of the secant line to the function $f(x)$ on the interval $[x_0, x_1]$. [2]
 - (f) Draw the graph of $f(x)$ together with the tangent lines and secant lines. [2]
2. (a) Roughly sketch the derivative graph of the following given function. [2]



- (b) Consider the function [5]
- $$f(x) = \begin{cases} 4 - x^2, & x > -1 \\ 2x + 5, & x \leq -1 \end{cases}$$
- (i) Sketch the graph of $f(x)$.
 - (ii) Determine whether the function $f(x)$ is continuous and differentiable at $x = -1$.
- (c) Find $\frac{dy}{dx}$, where $y = \cot^3 \sqrt{2x - 3 \sin x}$. [3]

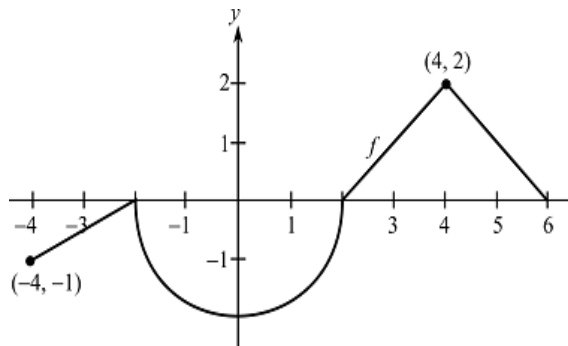
3. (a) Use appropriate formula of geometry to **evaluate** the following integrals: [5]

(i) $\int_{-3}^3 \{(x+4) - \sqrt{9-x^2}\} dx$ (ii) $\int_{-2}^3 (5 - |x+1|) dx$

- (b) The graph of $f(x)$ is shown. **Evaluate** the following definite integrals.

(i) $\int_{-2}^6 f(x) dx$ [3]

(ii) $\int_0^4 |f(x)| dx$ [2]



4. (a) Evaluate the following integrals: [6]

(i) $\int \frac{t^3 - t^2 \cos t - 5t + 1}{t^2} dt$

(ii) $\int \frac{x}{1+x^4} dx$

(iii) $\int x \sin 2x dx$

- (b) **Find and Sketch** the area of the region enclosed by the parabola $y = x^2$ and the line $y = x + 2$. [4]

BEST OF LUCK!!!